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Report Highlights:

Post forecasts a slight increase in fuel ethanol imports in 2026, driven by modest growth in the fuel pool following the declaration of energy emergency in the Philippines on March 24, 2026. Post forecasts a moderate increase in fuel ethanol production, supported by lower feedstock prices, particularly for molasses. Post forecasts both fuel ethanol and biodiesel consumption to expand in 2026, though at a slower pace, reflecting weaker vehicle sales and elevated fuel prices. The Philippines has no domestic sustainable aviation fuel (SAF) production, but efforts to develop the SAF industry gained further traction in 2026.

I. Executive Summary

Post forecasts fuel ethanol imports to increase 1 percent in 2026, driven by modest consumption growth as weaker vehicle sales constrain fuel pool expansion amid the national energy emergency. Fuel ethanol consumption is further moderated by only nine pump stations offering E20 and persistent concerns about vehicle compatibility, which continue to limit broader E20 uptake. Biodiesel consumption is similarly forecast to increase slightly by 1 percent. Sustained fuel price rollbacks could ease inflationary pressure and support a modest recovery in both gasoline and diesel consumption.

Vehicle sales remain a key driver of biofuels consumption. The automotive industry reported a month-over-month increase in sales in May 2026, following a double-digit percentage decline in units sold from January to April 2026 relative to the same period in 2025. Large-scale public transportation investments, such as metro rail expansion, and the government's electric vehicle policy are not expected to affect biofuels consumption in 2026.

Post forecasts fuel ethanol production to increase by 2 percent in 2026, supported by lower average molasses prices linked to higher total molasses supply, which contributed to a reduced bioethanol reference price (BRP). The Sugar Regulatory Administration publishes the BRP twice a month as a benchmark for the price negotiations of oil companies and bioethanol producers. Despite forecasted supply gains, feedstock availability may face downside risks in the years ahead, including the potential conversion of sugarcane lands to other uses. To address longer-term feedstock constraints, the industry is exploring alternative feedstocks such as corn, sweet sorghum, cassava, and nipa, though these remain in the research and development stage.

As of April 2026, 13 accredited bioethanol producers operate in the Philippines with a combined production capacity of 468 million liters per year (MLPY). One additional facility with a capacity of 48 MLPY is under construction. Once operational, total capacity will reach 513 MLPY. On the biodiesel side, 15 accredited producers hold a combined installed capacity of 907 MLPY, though one facility has been non-operational since 2025. Three additional facilities have received accreditation for construction, representing a combined potential capacity of 201 MLPY. Current overcapacity relative to the B3 mandate reflects industry expansion undertaken in anticipation of a policy shift to B5, consistent with the Philippine Energy Plan.

The Philippines has no domestic SAF production, but efforts to develop the SAF industry gained further traction in 2026. A multistakeholder working group was established to accelerate priority actions for developing the country's SAF industry. Several partnerships were also forged, including developing SAF from coconut-based feedstocks, and creating a platform for capacity building and knowledge exchange in sustainable aviation. The Philippine government also identified SAF as a measure under its fuel transition plan for the transport sector, though details of the plan have not yet been released.

II. Policy and Programs

The Philippines has enacted two laws to promote renewable energy (RE): the Biofuels Act of 2006 (RA 9367) and the Renewable Energy Act of 2008 (RA 9513), each with their respective implementing rules and regulations (IRR). For key features of these laws, see Biofuels Annual 2021. The Department of Energy (DOE) issued policies related to the Biofuels Act, primarily in the form of Department Circulars (DC). For the detailed list, see Biofuels Annual 2023.

The **Biofuels Act** seeks to reduce dependence on imported fuels by promoting the development and mandating the use of locally sourced biofuels, specifically bioethanol and biodiesel. Domestically produced bioethanol uses sugarcane (primarily molasses) as a feedstock, while biodiesel is produced from coconut oil transesterified into coco methyl ester (CME). Policy implementation focuses on production and consumption. No policies have been issued to incentivize reducing the carbon intensity (CI) of existing biofuels over time.

The **Renewable Energy Act** aims to accelerate the exploration and development of RE sources, including biomass, geothermal, solar, hydropower, ocean energy sources, and wind. The Act envisions the Philippines achieving energy self-sufficiency through sustainable development of RE resources and reducing reliance on fossil fuels, thereby minimizing vulnerability to international market price fluctuations. While the National Renewable Energy Program (NREP) does not cover transport biofuels, the Philippine Energy Plan (PEP) 2023-2050 includes biofuels in the energy mix.

A. Renewable Energy and Greenhouse Gas (GHG) Emissions

By signing the Paris Agreement, the Philippines committed to reducing carbon dioxide (CO_2) and other GHG emissions by scaling up renewable energy activities, including the use of biofuels. In April 2021, the Philippines submitted its first Nationally Determined Contributions (NDC), committing to a 75 percent reduction or avoidance of GHG carbon dioxide equivalent (CO_2e) emissions by 2030, relative to its business-as-usual scenario for 2020-2030. Reduction of CO_2e emissions will come from the energy, transport, waste, agriculture, and industry sectors. Several measures are expected to contribute to the NDC's fulfillment, including increasing the biofuels blends and advancing discussions on SAF. The Philippines has not updated its NDC commitments since April 2021.

Transforming the country's transport sector from fossil fuel-reliant to RE-dependent is a priority for the Philippines. Transport accounts for one-third of GHG emissions in the Philippines. Notably, fuel ethanol has half the carbon footprint of refined petroleum products, underscoring the potential role of biofuels in decarbonizing the transport sector. Post monitors three programs in this space: SAF, Public Utility Vehicle (PUV) Modernization, and Electric Vehicles (EV). For details of these initiatives, refer to subsections on SAF and Jet Fuel Market and Policy/Programs Impacting Rate of Growth in the Fuel Pool.

The government's EV initiatives are supported by the Electric Vehicle Industry Development Act (EVIDA), which lapsed into law in April 2022. EVIDA aligns with the country's policy to ensure energy security and independence by reducing reliance on imported fuel in the transportation sector. The law provides tax incentives for EV buyers and requires businesses to establish EV charging stations.

B. Policy and Programs on Biofuels

Policy Framework. The DOE leads the implementation of the Biofuels Act and its IRR, and prepares the National Biofuels Program consistent with the PEP. The National Biofuels Board (NBB) was established following the law as an oversight body composed of eight government entities (**Table 1**). The NBB acts as a recommendatory body on biofuels policies, assisted by a Technical Secretariat, whose powers and functions are monitoring implementation and recommending policies to be promulgated by the DOE.

Table 1. The National Biofuels Board

Chair: Secretary, Department of Energy (DOE)
Vice Chair (Bioethanol):
Administrator, Sugar Regulatory Administration (SRA)
Vice Chair (Biodiesel):
Administrator, Philippine Coconut Authority (PCA)
Members:
Secretaries, Department of Agriculture (DA),
Department of Trade and Industries (DTI)
Department of Science and Technology (DOST)
Department of Finance (DOF)
Department of Labor and Employment (DOLE)
Non-voting Members:
Secretaries, Department of Environment and Natural Resources (DENR)
Department of Transportation (DOTr)
Department of Agrarian Reform (DAR)
Chairman, National Commission of Indigenous Peoples (NCIP)

1. Fuel Ethanol and the Gasoline Market

Blend Mandates. The NBB determines the feasibility and recommends adjustment in the minimum mandated biofuel-blends subject to the availability of locally sourced biofuel. In compliance with the law, the 5 percent blend of fuel ethanol (E5) was mandated in the second year of the Biofuels Act 2009. The 10 percent blend (E10) was mandated in 2011. See also **Table 2**.

Table 2. Bioethanol Blends Implementation

Blend	Date Mandated and Policy	Date Implemented
Mandated		
E5	February 5, 2009 DC 2009-02-0002	February 12, 2009
E10	February 6, 2011 DC 2011-02-0001	February 27, 2011
Voluntary		
E20	May 7, 2024 DC 2024-05-0014	June 2024

Source: DOE

Discretionary E20 blend. On May 7, 2024, the DOE signed the Department Circular (DC) 2024-05-0014, which establishes the implementing guidelines and specifications for the voluntary roll-out of E20 gasoline. Under DC 2024-05-0014, oil companies can offer E20 to consumers on a voluntary basis, provided they comply with the approved standard. Fuel retailers providing E20 to consumers are directed to conspicuously display an "E20-Compatible Vehicle Advisory" to properly guide customers opting to use E20 for their vehicles (see also GAIN Report). The NBB approved discretionary E20 to reduce fuel pump prices and ease the burden of rising costs on consumers by offering a less expensive alternative gasoline supported by much cheaper imported fuel ethanol. This, in turn, would encourage more local investments. To date, one oil company offers E20 at nine fuel pump stations currently located in the country's capital.

Beyond the E10 mandate, the Philippines continues to study and explore a higher ethanol blend mandate (see also GAIN Report). The slow expansion in the number of pump stations offering E20, combined with concerns about vehicle compatibility, continues to constrain broader E20 uptake.

Fuel Quality Standards. The DOE promulgated the Philippine National Standard (PNS) for E20 on October 31, 2023 (**Table 3**). Oil companies blend fuel ethanol with gasoline to comply with the PNS, using appropriate blending methodologies at their refineries, depots, or blending facilities prior to selling ethanol blends. The DOE conducts semiannual monitoring and sampling activities to ensure compliance with the PNS.

Table 3. Bioethanol Quality Standards

Blend	PNS Number	Promulgation
E-100/E98	PNS/DOE QS 0017:2014	January 29, 2014
E10	PNS/DOE QS 008:2018	May 2018
E20	PNS/DOE QS 019:2023	October 31, 2023

Source: DOE

2. Biodiesel/Renewable Diesel and Diesel Markets

Blend Mandates. In 2024, following recommendations from the NBB, the DOE required fuel companies to implement a 3 percent CME blend in all diesel fuel on October 1, 2024, with annual 1 percent increases in 2025 and 2026 (**Table 4**). However, on May 29, 2025, the NBB suspended B4 implementation in 2025 and B5 in 2026 due to the elevated coconut oil prices in the international market. According to the World Bank, the price of coconut oil averaged $2,480/metric ton (MT) in 2025, up 63 percent from the 2024 average of $1,519/MT. From January to May 2026, international coconut oil prices averaged $2,276/MT, approximately 2 percent lower than the same period in 2025 ($2,315/MT). As of this writing, no announcement has been made regarding the implementation of the B4 and B5 mandates, limiting biodiesel consumption growth in the country.

Table 4. Biodiesel Blends Implementation

Blend	Date Mandated and Policy	Date Implemented
B1	May 17, 2007 DC 2007-05-006	June 7, 2007
B2	February 5, 2009 DC 2009-02-0002	February 12, 2009
B3	May 7, 2024 DC 2024-05-0013	October 1, 2024
B4	May 7, 2024 DC 2024-05-0014	
B5	May 7, 2024 DC 2024-05-0014	

Source: DOE

On March 24, 2026, President Ferdinand Marcos Jr. issued Executive Order (EO) No. 110, declaring a state of national energy emergency in response to the impact of the geopolitical conflict in the Middle East. The Philippines, which is heavily reliant on imported petroleum products, has faced significant spike in prices since the conflict began in late February. Under EO 110, the national energy emergency remains in effect for one year, unless extended or lifted by the President. The EO also mandates the creation of the Unified Package for Livelihoods, Industry, Food, and Transport (UPLIFT) Committee, chaired by the President, to ensure the stability of the domestic energy supply, the uninterrupted delivery of essential services, the continuity of economic activity, and the welfare of all citizens, as well as to mitigate the impact of the Middle East conflict. Specific measures implemented under UPLIFT are discussed in the Policy/Programs Impacting Rate of Growth in the Fuel Pool section of this report.

The national energy emergency prompted stakeholders to advocate for higher biodiesel blend mandates. A local multistakeholder committee on coconut under the Philippine Department of Agriculture urged an immediate transition to B5, arguing that a higher blend would reduce dependence on imported fuel and provide a stable market for Filipino coconut farmers.

A local biodiesel group similarly recommended increasing the blend to B5. According to the group, raising the mandate to B5 would create an immediate and significant market for over 3 million coconut farmers, absorb 30 percent of annual domestic coconut oil production, and displace approximately 430 million liters of diesel imports annually. The group further noted that while biodiesel adds a few pesos per liter, it improves fuel mileage by approximately 6 percent at the current B3 blend and by approximately 10 percent at B5, offering consumers good value for money.

Higher biodiesel blends also support the attainment of GHG emissions reduction targets under the Paris Agreement. A recent study found that coco-biodiesel emits approximately 78 percent less GHG than fossil diesel across its full life cycle, from field to tailpipe.

Meanwhile, Energy Secretary Sharon Garin indicated a fuel transition plan for the transportation sector to reduce the country's dependence on imported petroleum products. Measures being considered include an increase in the biodiesel blend to 50 percent. According to DOE, the transportation sector accounts for about 67 percent of the local oil demand.

Fuel Quality Standards. The DOE promulgated the PNS for B100 and B5 as early as 2015 in anticipation of increasing the blend from B2 to B5 in 2016 (**Table 5**). The B5 mandate did not happen because of the more lucrative export market for coconut oil. Manufacturers took advantage of the high price of coconut oil in the international market, reducing the available feedstock for CME. The NBB recommended implementing a gradual increase in biodiesel mandate, hence the development of the standards and promulgation of B3 and B4 PNS in 2021. Oil companies undertake the blending of diesel with CME in compliance with the PNS. The DOE conducts quarterly monitoring and sampling activities to ensure compliance with the PNS.

Table 5. Biodiesel Quality Standards

Blending Rate	PNS Number	Promulgation
B100	PNS/DOE QS 002:2015	November 27, 2015
B2 (automotive)	PNS/DOE QS 004:2017	December 18, 2017
B2 (industrial)	PNS/DOE QS 013:2017	December 18, 2017
B3	PNS/DOE QS 015:2021	December 27, 2021
B4	PNS/DOE QS 017:2021	December 27, 2021
B5	PNS/DOE QS 010:2015	November 27, 2015

Source: DOE

3. SAF and Jet Fuel Market

There is no SAF production in the Philippines. SAF, a sustainable alternative to fossil fuel-based aviation fuel, can be produced from plant-based feedstock such as forestry and agricultural waste, as well as used cooking oil. A study to assess the feasibility of local SAF production is being undertaken by DOTr, CAAP, and global aircraft manufacturer Airbus. The study, the completion of which remains undisclosed, will cover macroeconomic data analysis, SAF feedstocks and production pathways evaluation, financing assessment and policy support, and an action plan. The study will support policy development and is expected to encourage industry stakeholders to advance SAF production in the Philippines.

The Philippines is currently drafting its SAF Roadmap and recently formed a SAF Committee under the National Biofuels Board. The DOE leads the NBB-SAF Committee, members include CAAP, DOTr, DTI-Board of Investments (BOI), PCA, SRA, and DOST. The SAF roadmap is looking at four pathways – hydroprocessed esters and fatty acids (HEFA), alcohol-to-jet (AtJ) technology, power-to-liquid (PtL), and fermentation-based SAF. The SAF roadmap is yet to be finalized to craft the policies and framework that will help advance the development of SAF in the country. The Philippines initially targeted to finish the roadmap in mid-2024. To date, there has been no announcement yet on the publication of the SAF roadmap.

In 2025, aviation fuel accounted for 3 percent of total fuel demand in the Philippines (**Table 6**), driven primarily by the country's two major carriers – Philippine Airlines (PAL) and Cebu Pacific (CEB). PAL, the country's flag carrier, took delivery of its first Airbus A350-1000 in December 2025. According to Airbus, this aircraft is the first of nine A350-1000s PAL will receive under its fleet expansion program. The A350-1000 can operate on up to 50 percent SAF. Major airlines operating in the Philippines have been supporting adoption through partnerships, pilot initiatives, and sustainability commitments (see also GAIN Report).

Table 6. 2025 Fuel Products Demand

Fuel Type	Volume (In '000 Barrel)	Share to Total (In Percent)
Diesel	46,121	51.8
Gasoline	25,095	28.2
Fuel Oil	5,481	6.2
LPG	4,251	4.8
Aviation Fuel*	3,017	3.4
Bioethanol	2,788	3.1
Biodiesel	941	1.1
Kerosene	241	0.3
Others**	1,181	1.3
Total	89,116	100.0

Source: DOE
Note: *Jet A-1 and AvGas; **Includes naphtha and asphalt.

Amid the national energy emergency, DOE identified SAF as one of the measures under its fuel transition plan. The SAF Policy Development Workshop held in March 2026 formed a multistakeholder working group to accelerate priority actions for the development of the country's SAF industry, with CAAP, Island Skies Alliance (ISA), Boeing, the Subic-Clark Alliance for Development, and the National Aviation Academy of the Philippines as prospective members. Additionally, a Memorandum of Understanding (MOU) was formalized between the National Aviation Academy of the Philippines and ISA, establishing a platform for sustained capability-building and knowledge exchange in sustainable aviation.

During the SAF workshop, concerns on feedstock availability, production technology, and regulatory support, among others, were also discussed. The CAAP recognized that the transition could be facilitated by the country's agricultural sector, as waste materials from rice and coconut production could be used as aviation fuel. For instance, an estimated 20 million metric tons of rice straws are generated annually and considered waste, burned or left in open field due to its minimal commercial value.

In a separate initiative, ISA signed a MOU with PCA to develop SAF from coconut-based feedstocks. According to PCA, the partnership intends to utilize non-standard coconuts, commonly referred to as "reject" coconuts, or those that do not meet food-grade or export specifications.

The DOE acknowledges international initiatives such as the Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), established by the ICAO, which offers a harmonized way to reduce carbon emissions in the aviation industry. The Philippines as an ICAO member joined CORSIA in December 2018 through CAAP. Under CORSIA's timetable, participating ICAO member states pledged to comply with the carbon dioxide offsetting requirements by 2024 to 2026. Mandatory compliance is set for 2027 to 2035. Philippine airlines such as PAL and Cebu Pacific are mandated to implement CORSIA by 2027.

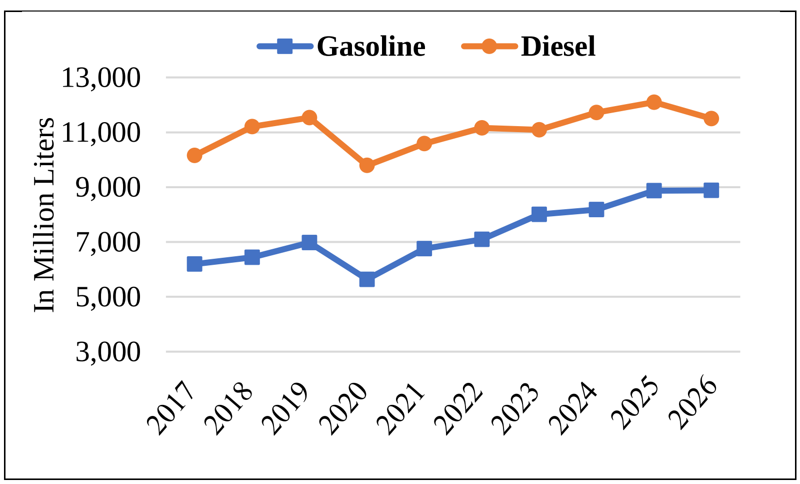
Financial Support for Biofuel Producers. To encourage investment, the government provides biofuels producers income tax breaks for the first seven years of operation, special realty tax rates on equipment and machinery, duty-free importation of equipment and machinery, zero-rated value added tax on purchases of goods and equipment, and zero percent value added tax rate on sale of bioethanol fuel, as stated in the RE Law of 2008.

In accordance with Section 17B of the Biofuels Act, the Social Amelioration and Welfare Program (SAWP) has been institutionalized wherein biofuel producers contribute a corresponding "lien" per liter of biodiesel and bioethanol produced and sold. Currently, the Department of Labor and Employment (DOLE) imposed a lien of PhP0.05 per liter of CME-based biodiesel, PhP0.19 per liter of sugarcane-based bioethanol, and PhP0.07 per liter of molasses-based bioethanol. Under the SAWP, qualified biofuel workers can take advantage of various assistance, such as for livelihood, training, education, social protection and welfare, and emergencies. The revised guidelines on the implementation on SAWP for biodiesel workers were issued June 3, 2021, while the revised SAWP for ethanol was issued in January 22, 2021, both with increasing social protection and benefits to workers. On February 22, 2024, the DOE issued Department Order No. 243-24 setting the amounts of the maternity and death benefit assistance programs for molasses-based bioethanol workers, pursuant to DOLE Department Order No. 222-21.

Policy/Programs Impacting Rate of Growth in the Fuel Pool

The Philippine fuel market is predominantly driven by gasoline and diesel (**Figure 1**). Consumption of both fuels has grown consistently since recovering from the COVID-19 pandemic. Growth is driven by the expansion of vehicle sales. According to data from the Chamber of Automotive Manufacturers of the Philippines (CAMPI) and the Truck Manufacturers Association (TMA), vehicle sales reached a record 491,395 units in 2025,

Figure 1. Growth in Gasoline and Diesel Pool, 2017-2026



Source: DOE and International Energy Agency (IEA) up nearly 4 percent from 473,842 units in 2024, approaching the industry's target of half a million units annually.

However, sales momentum slowed in 2026 amid rising fuel prices. CAMPI and TMA reported that vehicle sales from January to April 2026 totaled 132,867 units, down by 12 percent from 150,654 units during the same period in 2025. Sales showed a modest recovery in May 2026, with approximately 33,600 units sold, up nearly 4 percent from April. The industry now projects full-year 2026 sales could decline by 5-8 percent, an improvement over its earlier forecast of an 8-10 percent decline, reflecting the recent uptick in sales and fuel price rollbacks. This forecast correspondingly affects gasoline and diesel pool size, tempering consumption growth for both bioethanol and biodiesel in the country for 2026.

Unified Package for Livelihoods, Industry, Food, and Transport (UPLIFT). Following the declaration of state of national energy emergency under EO 110, the government implemented the UPLIFT program to mitigate impact of the fuel price spike across different sectors, including transport. By supporting the continuity of transport services amid the energy crisis, UPLIFT helps sustain fuel demand in the pool at a time when rising prices risk dampening overall fuel consumption. This measure partially offsets what could otherwise be a more significant contraction in the fuel pool.

One key intervention under UPLIFT is the fuel subsidy program. The government provides a PhP10 ($0.16)-per-liter fuel subsidy to eligible public utility jeepney (PUJ) and PUV express drivers through at least 900 fuel retail stations nationwide. As of May 29, 2026, the program had reached 673,249 recipients.

The government also implements the Service Contracting Program, which provides payments to drivers and operators and offers commuters at least a 20-percent fare discount. Program routes are designed to connect to train lines and major bus corridors, supporting efficient and reliable travel. The program targets 50,000 PUVs and 1,000 operators, with an expected benefit to 15 million passengers. As of May 18, 2026, 16,339 vehicles had enrolled. A separate 50 percent discount applies to railway transport, with 34.8 million passengers served as of May 17, 2026.

Additionally, the government launched the Libreng Sakay (Free Ride) Program along major arterial routes to provide commuters with an alternative amid transport service gaps, alleviate the impact of rising fuel prices, and ensure the continuous flow of public transport services particularly during peak rush hours and major holidays. President Ferdinand Marcos Jr. directed identified government departments, agencies, and instrumentalities to deploy available shuttle buses, coasters, and other transport assets to provide immediate relief to workers and ordinary commuters. As of May 31, 2026, the program had served 78,140 commuters.

EV Program. The government is implementing the Comprehensive Roadmap for the Electric Vehicle Industry (CREVI), which sets a short-term target of 2.45 million EVs and 66,500 electric vehicle charging stations (EVCS) from 2023 to 2028, in line with the clean energy scenario. The Philippines remains far from achieving these targets (**Table 7**). The total number of registered EVs reached nearly 61,000 units as of 2025, representing less than one percent of the total 14.8 million vehicles in the country based on the DOTr's data. At this level, EVs are not expected to have a measurable impact on fuel pool size in 2026. However, the IEA projects that the EV share of total car sales in the Philippines could rise to 45 percent by 2035. This trajectory, if realized, would exert downward pressure on fuel pool demand in the long-term.

Table 7. EV Industry Updates

Particulars	2023-2028 CREVI Target	Updates
No. of EVs	**2,454,200**	**60,906**
Cars (Sedan, SUVs, UVs)	553,000	59,223
Tricycles and Motorcycles	1,899,000	1,608
Bus	2,200	69
No. of EVCS	**66,500**	**1,600**

Source: DOTr and DOE (EV Industry Situationer)
Note: No. of EVs is as of 2025, while no. of EVCs is as of April 2026.

To eliminate gasoline-powered cars by 50 percent by 2040, the national governments laid the groundwork for a strong EV market by providing tax breaks for EV imports until 2028, and other incentives such as a production-linked incentive scheme, tax rebate or research and development, subsidy for EV purchase, and reduced electricity tariff for EV charging. The Department of Trade and Industry (DTI) set the policies for the manufacturing of EVs and issued incentives to encourage EV investment such as the removal of tariffs, excise duty exemptions, and VAT exemption for raw materials/parts/capital equipment to be used in EV manufacturing.

In January 2023, EO 12 was issued, temporarily modifying the rates of import duty on EVs, parts, and components, prompting importation and setting up of sales outlets or distributorships of mostly Chinese-made EVs. The bulk of EVs in the Philippines are e-tricycles and e-motorcycles, followed by electric utility vehicles like e-jeepneys.

PUV Modernization Program. Implemented by DOTr, the PUV Modernization Program emphasizes the environmental benefits of modernizing traditional jeepneys. Jeepneys have served as a primary mode of public transportation since the 1950s, originating from Willys Jeeps left behind by American troops after World War II. Over the decades, these customized, privately owned vehicles have become the backbone of the Philippines' public transit system. The program calls for the phase out of jeepneys that are at least 15 years old. Of the nearly 180,000 jeepneys that need to be modernized, the EV industry is targeting to supply 10 percent of this potential demand. However, the full compliance with the PUV Modernization Program has been extended. Progress to date remains limited due to high investment cost and lack of effective incentives. The slow pace of modernization has tempered the program's near-term impact on the fuel pool.

Large-scale, public transport investment programs. There are several large-scale investments in transportation infrastructure in the Philippines. These investments will eventually have an impact on the demand for biofuels in the country. According to the DOTr, the first underground railway system in the Philippines called the Metro Manila Subway Project (MMSP), is expected to be completed by 2032. The 33-kilometer (km) underground railway system has 8-car trainsets that can carry up to 2,200 passengers per train, running as fast as 80 km/hour, and train arriving at stations every 5 minutes. In 2024, the project has an estimated total cost of PhP488.5 billion ($8.79 billion). Aside from the MMSP, the Philippine railway sector infrastructure projects are listed below. According to DOTr, the commercial launch of MRT Line 7 and the North-South Commuter Railway in 2027 could replace up to 200,000 private vehicles, each carrying five passengers. A study further noted that DOTr's rail projects are estimated to contribute the most to GHG reduction, at 6.79 percent in 2030 and 4.23 percent in 2040.

- MRT Line 7, a 22-km long rail transit that will connect dense areas of northeast Metro Manila, which can accommodate 300,000-800,000 passengers daily.
- North-South Commuter Railway Project (NSCR) – a 147-kilometer elevated railway line that will connect Metro Manila in three regions.
- LRT 1 Cavite Extension Project – additional 11.7 kilometers extension of the existing LRT 1, which can accommodate additional 300,000 passengers, from LRT-1's 500,000 passengers daily.
- MRT Rehabilitation Project – involves the restoration of the existing MRT Line 3 to design condition. The rehabilitation is expected to restore capacity from 300,000 to 600,000 passengers per day.
- The MRT Line 4 and PNR South Long-Haul Project are in their pre-construction stage, while the Mindanao Railway Project Phase 1 is in project development phase.

C. Environmental Sustainability and Certification

The BOI extended a zero-VAT rating to the sale of raw materials used in biofuels production under the National Internal Revenue Code, as amended by the Expanded VAT Reform Law of 2006, in accordance with the Omnibus Investment Code of 1987 (Executive Order No. 226). BOI also provides incentives to registered biofuels manufacturers under the CREATE MORE Act, and the 2025 Philippines' Strategic Investment Priority Plan (SIPP) 2025-2028 which identifies projects under the Clean Energy as among the inclusions.

At present, biofuel plants are not required to certify the carbon intensity (CI) of their fuels on their own nor through third-party certification. Producers do not comply with any GHG sustainability requirements as there is no mandatory certification for biofuels placed in the domestic market. The lack of incentives lowers the drive of biofuel producers to bring down the CI of the fuels they market. Government policy is focused on compliance with the biofuels mandate and has not yet released guidance on the CI certification.

Table 8. Biofuels GHG Avoidance*

Year	Bioethanol (KTCO2e)	Biodiesel (KTCO2e)
2016	499.59	551.98
2017	516.61	516.05
2018	668.81	520.39
2019	782.92	551.53
2020	609.38	406.75
2021	788.81	484.74
2022	810.60	512.53
2023	843.11	575.91
2024	853.49	677.21
2025	826.80	886.27
2026**	284.54	278.03

Source: DOE
Note: *Formula of DOE-EPPB based on actual sales; **As of April 30, 2026.

In 2025, fuel ethanol used in the country resulted in a reduction of more nearly 827,000 tons of CO_2e in line with the efforts by the Philippine government to comply with its international commitment to reduce GHG emissions through RE (**Table 8**). Among the objectives of the Biofuels Act are a) develop and utilize indigenous renewable and "sustainably-sourced" clean energy sources to reduce dependence on imported oil; b) mitigate toxic and GHG emissions; and c) ensure the availability of alternative and renewable clean energy without any detriment to the natural ecosystem, biodiversity and food reserves of the country. The Philippine government investigates the realization of the sustainable development goals, wherein economic progress supports the protection of the environment and the health and safety of the people. The DOE-Energy Policy and Planning Bureau (EPPB) continues to monitor the GHG avoidance contribution of the actual local consumption of biofuels in the country. To date, there is no third-party certification of GHG values released by DOE-EPPB.

In 2019, the University of the Philippines Los Baños (UPLB) completed a study on "Life Cycle Assessment in terms of Carbon Debt and Payback Analyses, Carbon Savings and Energetics Studies of Biodiesel Production from Coconut Oil in the Philippines." The study calculated carbon intensity of Philippine biodiesel from crude coconut oil and refined coconut oil at 32.8 gCO2e/MJ and 31.5 gCO2e/MJ, respectively. However, under the current B2 blending requirement, this only results in a GHG reduction of 1.2-1.3 percent, which is still minimal (near zero) to date, though this factor would grow to 12.6-12.9 percent under a B20 scenario as earlier envisioned by DOE.

UPLB also conducted a life cycle assessment of fuel ethanol production, considering 10 out of 12 bioethanol distilleries during 2019-2020 (September to August). The study calculated the Philippines' overall ethanol carbon intensity at 46.8 gCO2e/MJ, albeit wide ranging from a 37 to 89 percent GHG reduction, depending on the ethanol plant observed. The study further estimated the adoption of an E20 mandate as earlier envisioned by DOE would result in four million tons of annual GHG savings by 2030.

The Biofuels Act exempts biofuel investors from wastewater fees but not from the obligation to secure a discharge permit. Under the Philippine Clean Water Act, the DENR enforced sustainability requirement on water quality, which requires facilities discharging regulated effluents to secure a permit to discharge (Section 14). The implementation of the Biofuels Act is linked to the provisions and in accordance with the objective of the Philippine Clean Air Act to develop and utilize cleaner alternative fuels. To date, no third-party certification is in place with permits only issued by the government.

There are no sustainability requirements for soil management or land use. The Philippines has no comprehensive land-use policy regarding cultivation of energy crops such as sugarcane for bioethanol or coconut for biodiesel. While the DA and DAR are mandated to determine the lands suitable to feedstock production, land use is largely determined by the local governments through ordinances. SRA monitors the sugarcane used for fuel ethanol production but there are no national standards or guidelines on how decisions are made on whether farmland is to be used for biofuel or food production.

D. Trade Policy, Import Duties/Licenses

The HS Codes for ethanol (excluding beverage ethanol) are 2207.10 undenatured and 2207.20 denatured. Currently, there is no import tariff on ethanol as tariffs fell to zero in 2016 (ASEAN) and 2020 (Most Favored Nation). A 1 percent duty is paid by oil companies on imported ethanol destined for gasoline blending. Biodiesel is harmonized under HS Code 3826.00 for pure biodiesel and biodiesel blends greater than 30 percent while petroleum oils containing 1-30 percent biodiesel by volume use HS Code 2710.20.

Fuel oil companies can import fuel ethanol only if domestic production cannot meet demand created by the mandate, which has been the case from the very earliest days of the fuel ethanol program. Sections

5.1 and 5.2 of the Biofuels Act recognized the country's insufficient feedstocks supply to meet the mandated volume of gasoline-fuel ethanol blend (e-gasoline). DOE and DOF issued DC 2006-08-0011 and Revenue Regulation No 8-2006, respectively, allowing oil companies to import fuel ethanol subject to the guidelines issued. The NBB determines the volume and allocates among fuel oil companies for compliance with the mandated blend.

By law, during the first four years of Biofuels Act implementation, in the event of supply shortage of locally produced fuel ethanol, oil companies were allowed to import fuel ethanol and benefit from a reduced tariff, but only to the extent of the shortage as determined by the NBB (Section 5 and 6 of the IRR, RA 9367). Due to insufficient feedstock, NBB still allows importation of fuel ethanol to comply with the mandated E10 blend requirement and discretionary E20 blend. The DOE promulgates the import guidelines in DC 2011-12-0013 requiring oil companies to purchase the entire monthly allocation of local fuel ethanol before they can import.

Allocation System. The bioethanol producers submit their committed volumes available for delivery to the DOE- Renewable Energy Management Bureau (REMB) on the first day of the allocating month, which is one quarter ahead of the delivery month. DOE-REMB endorses the consolidated monthly committed volumes for any given quarter to OIMB. The OIMB calculates and circulates the respective local monthly allocations (LMA) of each oil company on the 10th day of each allocating month based on the average market share. The LMA issuance serves as the regulatory mechanism for the exhaustion of local bioethanol. While imported ethanol fills in the supply gap, it is calculated based on gasoline demand. Importations of ethanol blended fossil fuels shall not be considered as part of the compliance with the LMA mandate. The quota allocation system ensures that imports do not displace locally produced fuel ethanol. DOE DC 2015-06-0007 provides revised guidelines on the utilization of locally-produced fuel ethanol. The revised DC omitted the notice of allowable bioethanol importation (NABI).

The Ethanol Producers Association of the Philippines (EPAP) requested the reinstatement of the NABI as stated in DC 2011-12-0013, which was omitted in the revised guidelines on the utilization of locally produced bioethanol in the production of e-gasoline. The DOE issued DC 2021-06-0014 or the revised circular for the accreditation and submission of notices and reports of the Philippine downstream oil industry (DOI) pursuant to Biofuels Act. The circular reinforces the DOE's mandate to strictly monitor the DOI. Additional information can be accessed here.

Diversion Prevention. The BIR issued Revenue Regulations No. 8-2006 prescribing the implementing guidelines on the taxation and monitoring of the raw materials used and the fuel ethanol-blended gasoline produced under the bioethanol program of the DOE. Imported fuel ethanol must be denatured by 2 percent gasoline in accordance with the formula prescribed under these regulations. This practice prevents the diversion of imported bioethanol to other usage other than fuel, i.e., potable alcohol. The denaturing of imported fuel ethanol must be conducted in the presence of authorized representatives of the oil industry participant, DOE, BIR and Bureau of Customs (BOC) within 48 hours immediately after completion of the unloading of the fuel ethanol from the foreign vessel and transfer thereof to the Customs-bonded storage tank. The REMB conducts monitoring and sampling activities to ensure compliance with the PNS – quarterly for biodiesel and semestral for bioethanol producers.

III. Ethanol

Table 9. Ethanol Used as Fuel and Other Industrial Chemicals (Million Liters)

Calendar Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026f
Beginning Stocks	Na	Na	25	14	14	10	18	22	16	18
Fuel Begin Stocks	Na	Na	25	14	14	10	18	22	16	18
Production	Na	Na	375	344	400	402	417	417	412	420
Fuel Production	235	297	346	280	355	375	387	382	377	385
Imports	407	439	432	408	487	505	584	589	696	725
Fuel Imports	276	285	257	241	225	277	246	409	485	490
Exports	0	0	0	2	5	0	20	0	0	0
Fuel Exports	0	0	0	0	0	0	0	0	0	0
Consumption	Na	Na	818	750	886	899	977	1,012	1,106	1,145
Fuel Consumption	511	557	614	521	584	644	629	797	860	875
Ending Stocks	Na	25	14	14	10	18	22	16	18	18
Fuel Ending Stocks	Na	25	14	14	10	18	22	16	18	18
Refineries Producing Fuel Ethanol (Million Liters)										
Number of Refineries	10	12	12	12	13	13	13	11	13	13
Nameplate Capacity	282	381	381	381	426	466	478	396	468	468
Capacity Use (%)	83.3%	78.0%	90.8%	73.5%	83.3%	80.5%	81.0%	96.5%	80.5%	82.3%
Co-product Production (1,000 MT)										
Bagasse	12	60	57	150	102	164	207	143	182	192
Feedstock Use for Fuel Ethanol (1,000 MT)										
Sugarcane	40	200	190	502	340	547	690	477	606	640
Molasses	950	1,080	1,240	1,076	1,340	1,344	1,265	1,297	1,253	1,300
Sugar	0	0	0	0	0	0	0	0	0	0
Market Penetration (Million Liters)										
Fuel Ethanol Use	511	557	614	521	584	644	725	797	860	875
Gasoline Pool 1/	6,199	6,441	6,973	5,636	6,757	7,091	8,008	8,174	8,873	8,882
Blend Rate (%)	8.2%	8.6%	8.8%	9.2%	8.6%	9.1%	9.1%	9.8%	9.7%	9.9%

Source: DOE, IEA, Bureau of Internal Revenue, Philippines Statistics Authority via Trade Data Monitor with Post estimates for 2026
Note: f = forecast
Na = neither government nor industry sources can provide stocks and production data from 2016 to 2018
1/= covers gasoline and all bio-components (ethanol)

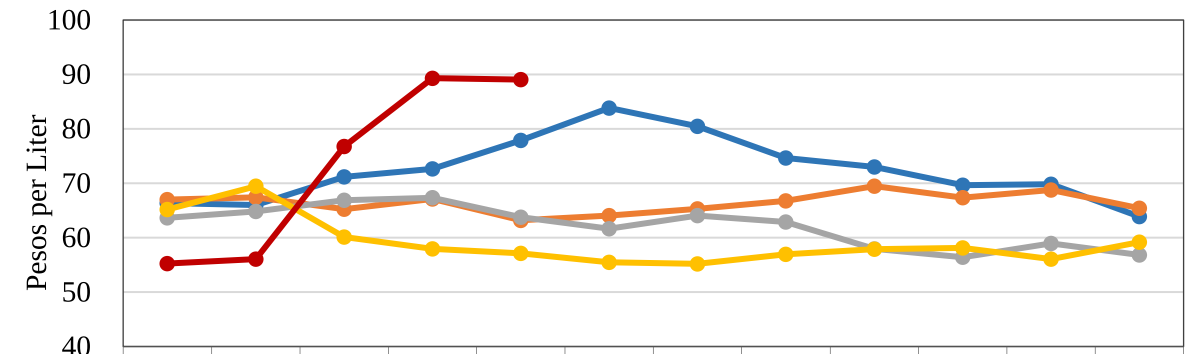
Consumption

Post forecasts fuel ethanol consumption at 875 million liters in 2026, up 2 percent from the revised consumption of 860 million liters in 2025. Growth is moderated by slower expansion of the fuel pool amid the Philippine national energy emergency declared on March 24, 2026, and limited uptake of voluntary E20 blending. The energy emergency will remain in effect for one year, unless extended or lifted by the President.

The moderate consumption growth outlook is primarily driven by weaker vehicle sales and softening gasoline demand. The automotive industry reported that vehicle sales in May 2026 increased 4 percent month-over-month, following a 12 percent decline in units sold from January to April 2026 relative to the same period in 2025. From March to May 2026, monthly gasoline prices averaged PhP85 ($1.36) per liter, 46 percent higher than the comparable period in 2025. Data from the Philippine Statistics Authority also show faster inflation averaging 4.5 percent as of May 2026, partially owing to elevated transport costs. Recent fuel price rollbacks could ease inflationary pressure and support a modest recovery in gasoline consumption.

Additionally, the number of pump stations offering E20 increased to nine, up from only two stations last year, marginally expanding fuel ethanol's share to the overall gasoline pool. More fuel retailers are expected to offer E20 at the pump, which would raise consumption especially among price-sensitive consumers. According to the fuel company offering E20, the price difference between E10 and E20 at their retail stations is approximately PhP2 (USD0.03) per liter. However, uncertainty about vehicle compatibility continues to temper E20 uptake, a concern that fuel retailers hope to address to broaden sales.

Figure 2. Monthly Average Gasoline Prices, 2022-2026



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2022	66.36	66.02	71.20	72.65	77.89	83.85	80.45	74.63	72.98	69.65	69.82	63.88
2023	67.01	67.44	65.23	67.14	63.21	64.05	65.32	66.8	69.48	67.33	68.78	65.43
2024	63.64	64.83	66.86	67.33	63.76	61.65	64.09	62.9	57.94	56.41	58.96	56.81
2025	65.19	69.45	60.12	57.93	57.13	55.48	55.20	56.95	57.89	58.15	56.09	59.21
2026	55.25	56.09	76.78	89.32	89.07							

Source: DOE
Note: Based on monitored retail outlets in the National Capital Region.

Production

Post forecasts fuel ethanol production at 385 million liters in 2026, up 2 percent from the revised 2025 production of 377 million liters. The forecast modest growth is supported by lower average molasses prices, which have contributed to reduced bioethanol reference price (BRP) for locally-produced ethanol (**Table 10**). According to SRA data, average millsite prices of molasses from October 2025 to May 2026 of the current marketing year (MY) averaged PhP9,658 per metric ton (MT), a 30 percent decline compared to PhP13,896/MT recorded during the same period in MY 2024-2025. The decline in molasses prices may be attributed to the 7 percent increase in total molasses supply, comprising carry-over stocks and local production. Approximately 80 percent of total domestic molasses supply is used as feedstock for ethanol production (see also Biofuels Annual 2025). To establish a level playing field, SRA releases the BRP twice a month as a benchmark for the price negotiations of oil companies and bioethanol producers.

Table 10. Molasses Price and BRP

Marketing Year*	Molasses Price (PhP/MT)	Sugar Composite Price (PhP/Lkg)	Bioethanol Price Index (PhP/Li)
2016-2017	8,544	1,430	53.68
2017-2018	6,364	1,440	50.96
2018-2019	9,694	1,532	57.40
2019-2020	11,828	1,484	61.07
2020-2021	9,352	1,500	57.07
2021-2022	11,735	2,014	65.79
2022-2023	14,510	3,163	82.07
2023-2024	15,700	2,578	80.56
2024-2025	13,932	2,679	80.26
2025-2026*	9,658**	2,298	65.25

Source: SRA
Note:
*For MY 2025-2026, the Philippine sugar marketing year runs from October of the preceding year to September of the current year. Prior to MY 2025–2026, the marketing year ran from September of the preceding calendar year to August of the current calendar year.
**As of May 2026.

Despite the current increase in total molasses supply, feedstock availability may face downside risks in the years ahead. Industry contacts in a key sugarcane-producing province in Luzon report that government agencies and private companies are engaging with sugar planters about coconut cultivation, amid the closure of a sugar mill in the area and opportunities in the coconut industry. The potential conversion of sugarcane lands into coconut plantations could gradually reduce molasses supply over time. Under the Biofuels Act of 2006, the importation of molasses for use as a biofuel feedstock is prohibited but allowed as feedstock in the production of denatured alcohol for industrial use (see also Biofuels Annual 2025). SRA issued policies for the moratorium on the importation of molasses until March 2026 due to the significant decline in molasse prices.

The industry is exploring corn and other potential feedstock, i.e. sweet sorghum, cassava, nipa and sweet potato, but these are still in the research stage and will take time to produce alternatives to sugarcane commercially for bioethanol processing, if they can achieve success at all. The DOE has projects on biofuels from nipa sap, a sugar-rich, nutrient-dense juice extracted from the stalk of the nipa palm, as feedstock and cellulosic ethanol production technology.

The PEP has set the targets for fuel ethanol production capacity on the assumption that all supply requirements are to be produced locally (see also Biofuels Annual 2022 for historical developments). In 2025, installed capacity of 468 million liters fell short of the 944 million liters target in PEP. As of April 2026, there are 13 accredited bioethanol producers in the Philippines, with a total production capacity of

468 million liters per year (MLPY). Additional capacity of 45 MLPY has been accredited for construction in Negros Occidental in the Visayas Region. Once all operational, total capacity will increase to 513 MLPY.

Meanwhile, imported molasses is allowed as feedstock for denatured alcohol production for industrial use. During the COVID-19 pandemic, DOE allowed bioethanol producers to repurpose their facilities for medical-grade alcohol production, with five producers shifting to medical-grade or rubbing alcohol to address domestic supply shortages. Denatured alcohol production is expected to continue growing, driven by sustained post-pandemic demand across the health sector and for personal care, household cleaning, and sanitation products.

Post revised downward its 2025 estimate for ethanol production, considering the actual production data from DOE.

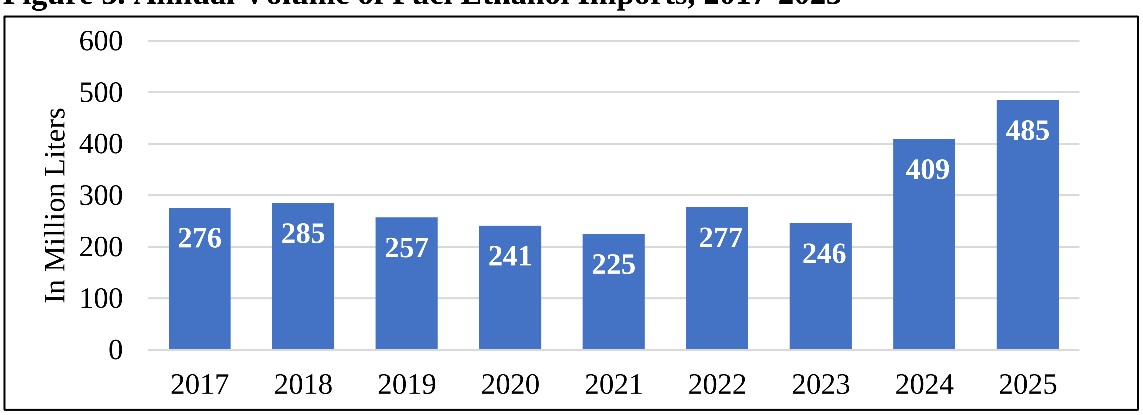
Trade

Post forecasts fuel ethanol imports to slightly increase to 490 million liters in 2026 from 485 million liters in 2025, driven by a modest growth in consumption amid the slow uptake of discretionary E20 blend. While local ethanol remains the priority, imports are expected to cover approximately 56 percent of total supply needed to meet both the mandated E10 and voluntary E20 blends. In 2025, imports also accounted for 56 percent of the total demand for fuel ethanol. Post revised upward 2025 imports data based on actual DOE data.

The United States has been the largest supplier of imported fuel ethanol, accounting for 89 percent of total fuel ethanol imports in 2025, up from 70 percent in 2024 (**Figures 3 and 4**). The remaining imports were sourced from neighboring Asian countries, particularly Thailand and China, displacing traditional supplier Brazil. Industry contact cites that the bioethanol producers are sourcing ethanol from Thailand, where excess supply has emerged following a slowdown in domestic demand. The demand slowdown is partly attributed to the government's reduction of subsidies for higher ethanol blends.

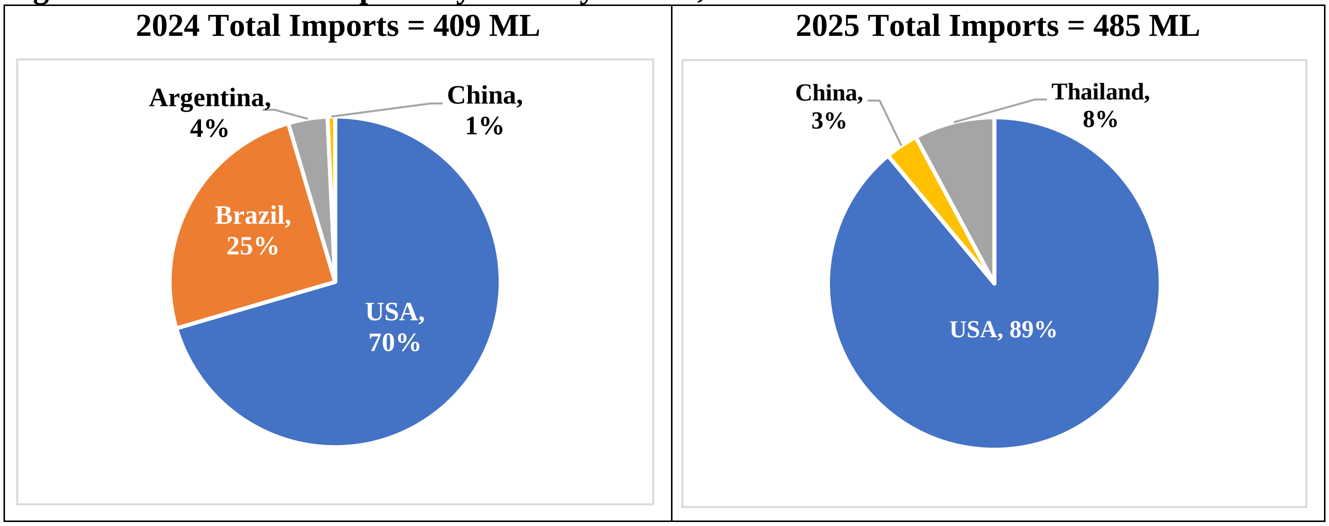
Post corrected total ethanol trade for previous years, considering the actual imports and exports data from the Philippines Statistics Authority accessed through Trade Data Monitor.

Figure 3. Annual Volume of Fuel Ethanol Imports, 2017-2025



Source: DOE

Figure 4. Fuel Ethanol Imports by Country Source, 2024 and 2025



Source: DOE

As of 2025, there were 23 accredited importers of fuel ethanol composed of the top fuel oil companies (Petron, Shell, Chevron) and fuel traders who supply small players.

Table 11. Accredited Downstream Oil Industry Biofuel Participants

Petron Corporation	. Insular Oil Corporation
Pilipinas Shell Petroleum Corporation	. Warbucks Industries Corporation
Chevron Philippines, Inc.	. Warbucks Southern Corporation
PTT Philippines Corporation	. ERA 1 Petroleum Corporation
Seaoil Philippines, Inc.	. Power Fill International Subic Inc.
Unioil Petroleum Philippines, Inc.	. Vmaximus Subic FreePort Corp.
Jetti Petroleum Inc.	. APEX Petroleum OPC
Phoenix Petroleum Philippines, Inc.	. Trafigura Philippines Inc.
Filoil Logistics Corporation	. Goldenshare Commerce & Trading, Inc.
Marubeni Philippines Corporation	. South Brookville Trading Corp.
Micro Dragon Petroleum, Inc.	. Felcor Petroleum Depot Corporation
High Glory Subic International Logistics, Inc.	

Source: DOE

IV. Biodiesel

Table 12. Biodiesel (Million Liters)

Calendar Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026f
Beginning Stocks	41	57	53	64	56	47	48	11	10	12
Production	220	220	242	188	198	203	225	266	348	370
Imports	0	0	0	0	0	0	0	0	0	0
Exports	0	0	0	0	0	0	0	0	0	0
Consumption	204	224	231	196	207	202	262	267	346	350
Ending Stocks	57	53	64	56	47	48	11	10	12	32
Production Capacity (Million Liters)										
Number of Biorefineries	11	11	12	13	13	13	13	13	13	15
Nameplate Capacity	575	575	608	708	708	678	678	791	791	907
Capacity Use (%)	38.3%	38.3%	39.8%	26.6%	28.0%	30.0%	33.2%	33.6%	44.0%	40.8%
Feedstock Use (1,000 MT)										
Coconut Oil	202	202	222	172	182	187	207	244	320	340
Market Penetration (Million Liters)										
Biodiesel	204	224	231	196	207	202	262	267	346	350
Diesel Pool, on-road use	8,086	8,200	8,413	6,440	7,460	8,191	8,100	7,641	7,756	7,780
Blend Rate (%)	2.0%	2.0%	2.0%	2.0%	2.0%	1.8%	2.3%	2.3%	2.9%	3.0%
Diesel Pool, total 1/	10,159	11,207	11,534	9,786	10,590	11,164	11,090	11,719	12,097	11,499

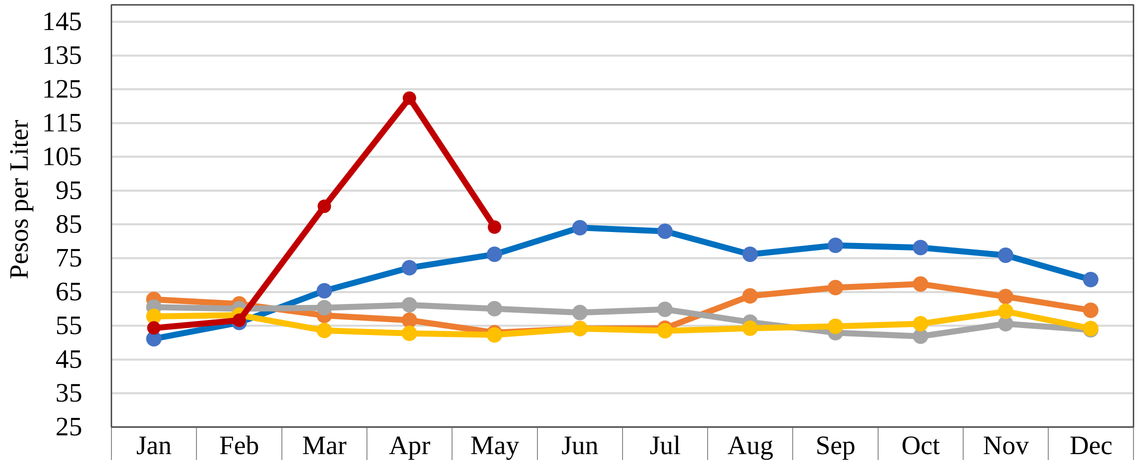
Source: DOE, IEA with Post estimates for 2026
Note: f = forecast
1/ = Diesel pools include all blended biodiesel.

Consumption

Post forecasts for biodiesel consumption to reach 350 million liters in 2026, a 1 percent increase over the previous year. This modest expansion is facilitated by the continuous implementation of the B3 mandate, but tempered by a potential decline in the diesel pool amid the national energy emergency. Post revised 2025 consumption downward, considering adjustments in actual stocks and production.

Consumption growth in 2026 has been partially offset by a surge in diesel prices stemming from energy supply disruptions tied to the fuel crisis. Diesel prices rose sharply, reaching PhP120 ($2.0) per liter in April 2026, the highest level in five years. To cushion the impact of the price spike, the Philippine government extended fuel subsidies to PUJ and PUV drivers, provided free rides, and offered discounted fares under the UPLIFT Program, ensuring continued transport services for Filipino commuters. If recent fuel price rollbacks are sustained, a recovery in vehicle activity could support additional vehicle sales and boost consumption going forward.

Figure 5. Monthly Average Prices of Diesel, 2022-2026



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2022	51.15	55.99	65.33	72.19	76.10	84.04	82.98	76.14	78.75	78.17	75.87	68.67
2023	62.79	61.45	58.05	56.58	52.96	54.17	54.21	63.78	66.29	67.36	63.63	59.54
2024	60.49	60.01	60.34	61.14	60.06	58.87	59.81	56.07	52.94	51.86	55.56	53.74
2025	57.79	58.15	53.62	52.77	52.3	54.14	53.46	54.24	54.86	55.57	59.23	54.11
2026	54.31	56.65	90.34	122.42	84.15							

Source: DOE
Note: Based on monitored retail outlets in the National Capital Region.

Price trend analysis over the years indicates that raising the biodiesel blend rate from B2 to B5 would result in higher pump prices in the Philippines relative to fossil diesel. However, this price effect is relatively modest at low-blend rates and has not deterred PCA from supporting the increase in the biodiesel blend mandate in support of the livelihood of around 2.5 million coconut farmers. An assessment of the broader effects on the coconut oil industry and a calculation of the expected diesel price increases would be warranted. That said, at low-blend rates, any price increase in the blended fuel over pure fossil diesel is largely 'washed-out' by the wide swings in oil and fossil diesel prices on the international market seen year after year. **By law, importation of biodiesel is not allowed, which could be the solution to manage the increase in diesel pump prices.**

Table 13. Prices of Diesel, CME and CNO

Year	Diesel Price (a) (PhP/Li)	Biodiesel (CME) Price Range (b) (PhP/Li)	Crude CNO Local Price (c) (PhP/Kg)
2017	31.78	45 – 92	90.41
2018	42.16	40 – 91	79.97
2019	41.60	35 – 70	51.73
2020	34.71	35 – 75	49.19
2021	42.16	58 – 90	97.16
2022	72.12 (b)	68 – 96	82.37
2023	60.07 (b)	53 – 62	65.97
2024	57.57 (b)	72.73*	75.08
2025	55.02 (b)		138.07

Source: (a) ADB Key Indicators Database Philippines
(b) DOE
(c) United Coconut Association of the Philippines
Note: *Average price

Production

Post forecasts biodiesel production at 370 million liters in 2026, up 6 percent from revised production level in 2025. This growth is supported by rising consumption under the current B3 mandate, and the increase in the number of biodiesel producers from 13 in 2025 to 15 this year. According to industry contacts, existing biodiesel plants can support blends up to B7, suggesting that B5 implementation could be accommodated by current production infrastructure. Industry contacts also note that while the bulk of coconut oil continues to flow to traditional export markets, the local biodiesel industry offers coconut farmers a more predictable market, an advantage that would become more pronounced with the implementation of higher blend mandates.

Post revised downward the biodiesel production in 2025 to 348 million liters, considering actual data from DOE. Post previously estimated a significant increase in production for 2025, with the assumption that B4 mandate would be implemented from October to December 2025. However, the NBB suspended its implementation, reducing actual 2025 production (see also Biofuels Annual 2025).

As of April 2026, 15 accredited biodiesel producers operate in the Philippines with combined installed capacity of 907 MLPY. However, one facility has been non-operational since 2025, due to expired accreditation, effectively reducing total capacity to 883 MLPY. Three facilities have received accreditation for construction, with a combined potential capacity of 201 MLPY. Current overcapacity relative to the B3 mandate reflects industry expansion undertaken in anticipation of a shift to B5, as guided by the PEP.

Raising the biodiesel blend mandate would require sustained development of domestic feedstock sources to ensure adequate supply. Research and development of biofuel production and utilization falls under the mandate of DOST. In December 2019, the DOST- Industrial Technology and Development Institute (DOST-ITDI) completed the study entitled Characterization/Performance Testing of the Biodiesel/Diesel Blends from Combined Feedstock of Various Vegetable and Used Cooking Oils. Biodiesel was produced from various feedstocks, namely: refined palm oil, used cooking oil, and rubber seed oil through optimized processes. Results of the analyses showed methyl esters from palm oil, used cooking oil, and rubber seed oil can be blended with CME for use as fuel additive to petroleum diesel.

Trade

No trade is allowed under the Biofuels Act, although imported palm-oil biodiesel could offset the expected price increase of a higher blend mandate through lower-cost imported biodiesel.

V. Advanced Biofuels

The DOST spearheads initiatives on biofuels research and development (R&D). DOST has recently funded the following projects on SAF development and production:

- Advancing Sustainable Aviation Fuel in the Philippines through the Establishment of a Demo-Scale Sustainable Aviation Fuel Production Facility Utilizing Used Cooking Oil as Feedstock
  This project is funded under the DOE's Renewable Energy Trust Fund (RETF) under the 2024 RESTI Call for Proposals. The general objective is to demonstrate CORSIA eligible SAF

production from used cooking oil via the HEFA process and assess its technical performance, production economics, and logistics cost.

- Advancing Sustainable Aviation Fuel in the Philippines: Propelling Green Energy Solution for Aviation Decarbonization. The general objective is to assess the techno-economic and environmental viability of SAF production in the Philippines from alternative feedstocks such as coconut oil, palm oil, and used cooking oil.

The Philippines also focuses on finding new feedstocks to produce more biofuels. The National Bioenergy Research and Innovation Center (NBERIC) was established in 2018 to realize R&D projects in bioenergy. NBERIC collaborates with various institutions on bioethanol R&D to maximize the NBERIC facility located at the Mariano Marcos State University (MMSU) in Batac, Ilocos Norte in northern Philippines. The USAID - Science, Technology, Research and Innovation for Development (STRIDE) program was instrumental in the establishment of NBERIC. The U.S. Grains and Bioproducts Council (USGBC) signed a Memorandum of Understanding with MMSU in July 2022 to establish a cooperative partnership, which facilitates the exchange of expert knowledge, technical information, and best practices as they relate to the biofuels industry and policy development.

The Pampanga State Agricultural University established the Bio-Energy Laboratory in 2017-2019 to determine the potential of the enormous volume of agricultural resources, products and wastes in Central Luzon as biofuels. The laboratory focuses on the thermal conversion and characterization of biomass used of agricultural wastes and other lesser studied agricultural resources.

The DOST-Philippine Council for Industry, Energy, and Emerging Technology Research and Development and the Industrial Technology Development Institute (ITDI) conducted R&D on biofuels in collaboration with other institutions such as DOE, DA, PCAARRD, and academia, particularly the UPLB and UP Visayas.

Bio-oil production from agricultural waste (2014-2020). Pyrolysis was conducted on a corn stover using a prefabricated reactor. Optimum conditions on temperature, time, and catalyst have yielded corresponding improvements in bio-oil and char yields. Recommendations for further studies include scaling up of equipment and facilities to support bio-oil production from other agricultural waste sources, working on higher pyrolysis temperature, and economic analysis to determine cost efficiency of the bio-oil and char.

Microalgae experiments were done by DOST and UP Visayas at the laboratory stage but the economic feasibility for commercial-scale production must be established. Likewise, studies on used vegetable oils, particularly from fast-food chains, were conducted by DOST, and more studies are needed on raw material quality control, and process optimization, since the resulting biodiesel had difficulty meeting standards. For bioethanol, DOST and UPLB have undertaken studies on sweet sorghum, and the economic feasibility of its cultivation vis-à-vis sugarcane needs to be ascertained.

A project funded by the NBB was completed by the ITDI in 2018 on multi-feedstock (coconut oil, palm oil, used oil, rubber oil) production of biodiesel, and further studies are needed for engine testing. These initiatives focused on viability of feedstocks and may not be considered advanced technology as has been solicited.

Appendix: Data Sources

Feedstock-to-Biofuel Conversion Rates

Biodiesel

Coconut oil	1 MT = 1,090 liters of biodiesel

Trade Data

Total ethanol trade (imports and exports) data in this report is sourced from the Philippines Statistics Authority via Trade Data Monitor, using HS codes 2207.10 and 2207.20. Fuel ethanol import data are sourced from the Philippine DOE.

Other Data Sources

Fuel ethanol and biodiesel production estimates are sourced from the DOE, along with stocks, fuel consumption, and feedstock use for previous years. For total ethanol production figures, Post also coordinates with the Bureau of Internal Revenue for available denatured alcohol data.

Gasoline and diesel demand data for 2017-2025 are sourced from the DOE, and IEA data is used to forecast 2026.

Attachments:

No Attachments